

Pipeline Before vs After

Hallucination Detection in RAG-based Interview Analysis
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This document summarises every architectural change made to the thesis pipeline, with the old approach on the left (red) and the new approach on the right (green). Each change is grounded in a peer-reviewed paper.

1. Model Registry

The model lineup changed from 6 models (including a gated LLaMA + Mistral-24B via API) to 7 models (all 7-8B for size parity, adding Qwen2.5-14B for scale ablation and Mixtral-8x7B for MoE architecture comparison).

BEFORE	AFTER
6 models: gpt-4o-mini, mistral-small-24b (Together.ai API), qwen, llama (gated), geitje, aya23	7 models: gpt-4o-mini, mistral-7B-Instruct-v0.3 (HF), qwen7B, qwen14B (4-bit), geitje, aya23, mixtral-8x7B (4-bit)
Mistral: Mistral-Small-24B via Together.ai API Size: 24B (violates parity)	Mistral: Mistral-7B-Instruct-v0.3 via HuggingFace (public, no token) Size: 7B (parity maintained)
LLaMA-3.1-8B included (requires HF access approval)	LLaMA removed (access pending). Replaced by Mixtral-8x7B (MoE) and Qwen2.5-14B (scale ablation)

Papers: Jiang et al. 2023 (arXiv:2310.06825) — Mistral-7B. Hui et al. 2024 (arXiv:2412.15115) — Qwen2.5-14B. Jiang et al. 2024 (arXiv:2401.04088) — Mixtral MoE.

2. Embedding Strategy

The old embedder (MiniLM) was designed for sentence similarity, not retrieval. BGE-M3 provides unified dense+sparse+ColBERT retrieval with +48% improvement on Dutch passages. A cross-encoder reranker is added as a second stage.

BEFORE	AFTER
Model: paraphrase-multilingual- MiniLM-L12-v2 Task: sentence similarity Retrieval: cosine only	Model: BAAI/bge-m3 Task: unified retrieval Stages: dense + sparse + ColBERT Dutch improvement: +48%
Single-stage retrieval: bi-encoder → top-3 final	Two-stage retrieval: bi-encoder → top-20 candidates cross-encoder reranks → top-3 final
Reranker: none	Reranker: BAAI/bge-reranker-v2-m3 (CrossEncoder, Nogueira & Cho 2019)

Papers: Chen et al. 2024 (arXiv:2402.03216) — BGE-M3. Nogueira & Cho 2019 (arXiv:1901.04085) — Cross-encoder reranking.

3. Chunking Strategy

Fixed-size character chunking cuts mid-turn, destroying speaker attribution context. Speaker-aware chunking preserves complete speaker turns, which is critical for interview data.

BEFORE	AFTER
Method: fixed 1000-char chunks No speaker boundary awareness Result: cuts mid-sentence, mid-speaker-turn	Method: speaker_aware_chunk() Splits at: Interviewer: Participant: Target: 400 tokens, 50-token overlap
Chunk boundary: arbitrary character position	Chunk boundary: regex pattern (?=(?:Interviewer Participant Agent Speaker\d*):)

Paper: Shi et al. 2023 (arXiv:2309.15127) — retrieval precision degrades when chunks split conversational context.

4. System Prompt

The old prompt was generic and did not force evidence citation. The new evidence CoT prompt requires the model to explicitly quote the transcript before answering, grounding every claim.

BEFORE	AFTER
style: "detailed" "You are an expert research assistant. Answer using the provided context."	style: "evidence_cot" STEP 1: QUOTE relevant sections STEP 2: VERIFY quote supports query STEP 3: ANSWER from verified quotes only
No explicit grounding requirement	Forced evidence citation Reduces BASELESS_INFO and EVIDENT_CONFLICT hallucinations

Papers: Wei et al. 2022 (arXiv:2201.11903) — Chain-of-Thought prompting. Yao et al. 2023 (arXiv:2302.00093) — ReAct grounded generation.

5. Hallucination Taxonomy

Two low-prevalence types removed (SPEAKER_MISATtribution 1.6%, TEMPORAL_CONFUSION 2.3%). One novel type added: ROLE_ATTRIBUTION_DRIFT — a sequential positional pattern not captured by any existing benchmark.

BEFORE	AFTER
7 types: 1. EVIDENT_CONFLICT 2. SUBTLE_CONFLICT 3. BASELESS_INFO 4. SPEAKER_MISATtribution 5. TEMPORAL_CONFUSION 6. SENTIMENT_MISREPRESENTATION 7. REFUSAL_HALLUCINATION	6 types: 1. EVIDENT_CONFLICT 2. SUBTLE_CONFLICT 3. BASELESS_INFO 4. SENTIMENT_MISREPRESENTATION 5. REFUSAL_HALLUCINATION 6. ROLE_ATTRIBUTION_DRIFT (NEW)
SPEAKER_MISATtribution: 1.6% pilot prevalence. Removed. TEMPORAL_CONFUSION: 2.3% pilot prevalence. Removed.	ROLE_ATTRIBUTION_DRIFT: Sequential positional pattern. 3+ latter-half sentences drift to wrong speaker. Grounded in positional decay theory.

Paper: Shi et al. 2023 (arXiv:2108.12409) — 'Lost in the Middle': positional decay. Gap papers confirm no existing benchmark captures this type.

6. Correction Loop

The original correction loop had two critical problems: Windows hardcoded paths (D:\thesis_hallucination\...) and a single correction pass. The new version uses dynamic path resolution and iterates up to 2 rounds.

BEFORE	AFTER
Paths: hardcoded Windows D:\thesis_hallucination\.env D:\RAG_THESIS\output\ Only works on dev machine	Paths: dynamic Linux _PROJECT_ROOT = Path(__file__).parent.parent.parent Works on ALICE HPC

Correction: single pass 1 round per hallucinated response (no early exit)	Correction: iterative max_rounds=2 Early exit if FAITHFUL per_round_faithfulness tracked
Output: D:\RAG_THESIS\output\correction_results.json	Output: results/correction_loop/correction_results.json

Papers: [Madaan et al. 2023 \(arXiv:2303.17651\)](#) — Self-Refine iterative refinement. [Shinn et al. 2023 \(arXiv:2303.11366\)](#) — Reflexion verbal reinforcement.

7. Strategy Comparison Summary Table

Complete overview of all architectural changes, their motivation, and expected impact.

Approach	Old Method	New Method	Paper Backing	Expected Improvement
Model lineup	6 models incl. Mistral-24B API, LLaMA gated	7 models, all 7-8B HF, add Qwen14B + Mixtral	Jiang 2023, Hui 2024, Jiang 2024	Size parity; scale + MoE ablation
Embedder	MiniLM (similarity)	BGE-M3 (retrieval, multilingual)	Chen et al. 2024 (arXiv:2402.03216)	+48% Dutch retrieval precision
Reranker	None	bge-reranker-v2-m3 cross-encoder	Nogueira & Cho 2019 (arXiv:1901.04085)	Better top-3 precision from top-20
Chunking	Fixed 1000-char	Speaker-aware 400-token	Shi et al. 2023 (arXiv:2309.15127)	Preserve speaker turns
System prompt	Generic detailed	Evidence CoT (3-step quote-verify-answer)	Wei 2022, Yao 2023	Reduce BASELESS_INFO, EVIDENT_CONFLICT
Hallucination taxonomy	7 types (incl. low-signal)	6 types + novel ROLE_AT TRIBUTION_DRIFT	Shi et al. 2023 (arXiv:2108.12409)	Novel contribution, 0 gap coverage
Correction loop	Single-pass, Windows paths	Iterative 2-round, Linux paths	Madaan 2023, Shinn 2023	Higher faithfulness gain per response
Inference backend	Transformers (sequential)	vLLM option (PagedAttention batched)	Kwon et al. 2023 (arXiv:2309.06180)	2-4x speedup, 12h → 2-4h ALICE
Fine-tuning	None	QLoRA rank-16 on FAITHFUL examples	Dettmers 2023, Hu 2021, Wang 2024	Target: hallucination rate < 5%